

BharatTrip Refund Escalations: Business Case & Solution Proposal

Executive Summary (TL;DR)

BharatTrip is experiencing a spike in customer/agent refund escalations despite stable refund volumes. Our analysis identified a severe operational disconnect between Support and Finance, leading to **100 dropped tickets** and **149 short payments**. By implementing a **Single Source of Truth (SSOT)** and an **AI-Driven Data Reconciliation Workflow**, we deploy a "Digital Workforce." This allows existing teams to eliminate manual data leakage and automate agent communications, driving a projected **85%+ drop in escalations** over the next quarter **without adding a single headcount**.

1. Defining the Problem

Customer and agent escalations regarding refunds are rising. The core underlying problem is a **severe operational disconnect between the Support and Finance teams**, characterized by siloed tracking systems and inconsistent definitions of refund statuses.

Key Communication Failures:

1. **Data Leakage:** Informal requests via WhatsApp and Email bypass the tracker altogether, meaning they are never reviewed or processed.
 2. **"Closed" Status Ambiguity:** The Support team marks tickets as "Closed" once handed off to Finance, misleading the agent into believing the refund has been disbursed.
 3. **Hidden Policy Deductions:** Finance applies necessary deductions (e.g., cancellation fees) and processes payouts, but this information is never relayed back to the agent by Support, causing "short payment" escalations.
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2. Data Analysis & Insights

By cross-referencing the `Support_Tracker`, `Finance_Tracker`, and `Escalations` logs via a deep database join, we identified the specific patterns driving the **155 total escalations**.

The Discrepancy Breakdown

Escalation Driver	Ticket Volume	Description
Dropped Handoffs	100	Logged by Support, but entirely missing from Finance tracking.
Deduction Mismatches	149	Amount promised by Support differs from amount disbursed by Finance.
Off-Tracker Leakage	24	Escalations with "no ref" originating from unlogged WhatsApp/Email.
Asynchronous Closure	47	Processed by Finance but missing from Support.

Insight: While a naive net volume calculation (755 Support - 689 Finance) suggests only 66 missing tickets, our deep ID-to-ID database reconciliation reveals the true scale of the failure is **100 dropped tickets**, masked by duplicate records and asynchronous closures.

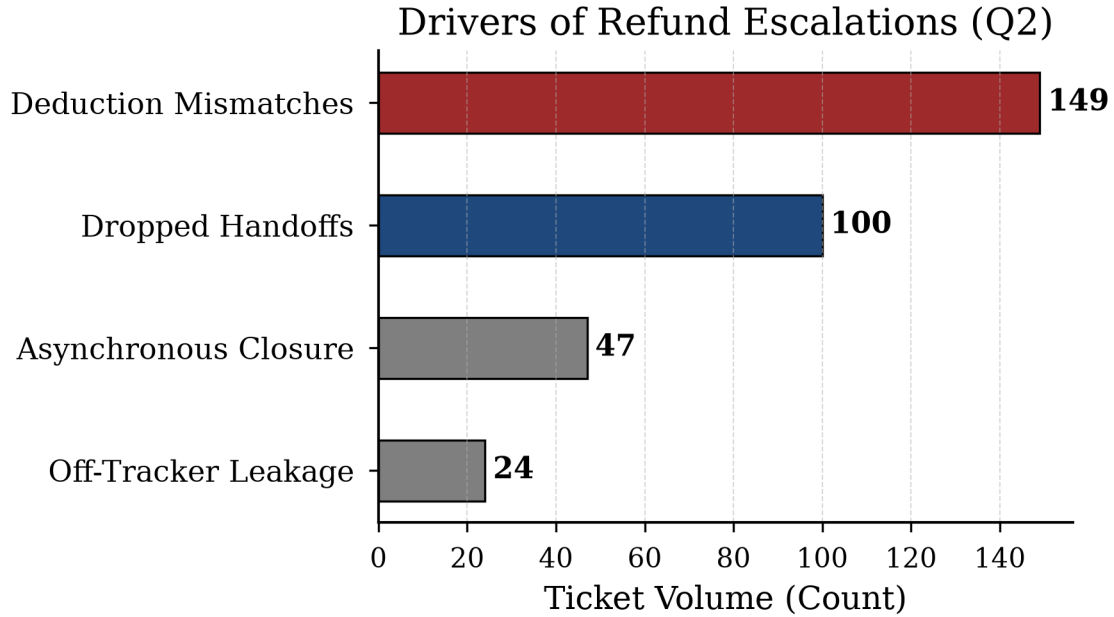


Figure 1: Metrics Data Leakage Visualization

3. Designing a Zero-Headcount Solution

To permanently bring escalations down without adding headcount, BharatTrip must establish a unified **Single Source of Truth (SSOT)** and automate the communication loop.

The Proposed Workflow (Digital Workforce):

1. **Unified SSOT Tracking:** Migrate both teams onto a single shared database.
2. **AI-Powered Ingestion:** Deploy an AI agent to monitor Support Email and WhatsApp. The AI acts as a digital clerk, parsing unstructured messages and logging them as structured tickets, eliminating “no ref” leakage entirely.
3. **Automated Reconciliation:** A system trigger automatically cross-references Finance payouts with Support quotes. If a deduction occurs, the AI drafts an explanatory email for the agent, preventing short-payment shock.

4. The AI Prototype: Advanced Multi-Agent Architecture

To deliver a production-ready solution, I developed `app.py`, a **Streamlit Web Application** showcasing an advanced Multi-Agent architecture powered by `gemini-3.5-flash`.

Operational Safety: Risks & Mitigations

Deploying AI in financial workflows carries inherent risks (e.g., hallucinated refund amounts or improper authorizations). Our architecture mitigates this using a **Human-In-The-Loop (HITL) Design**:

- **Financial Compliance:** The AI is strictly limited to *drafting* explanatory emails and structuring incoming data. It cannot authorize payouts. A human Support representative must click “Approve & Send” on the AI’s drafted email before any external communication occurs.
- **Secure Integration (MCP):** In production, the LLM will bridge with the internal SSOT database via the **Model Context Protocol (MCP)**. This guarantees that raw database credentials are never

Proposed SSOT & Multi-Agent Architecture

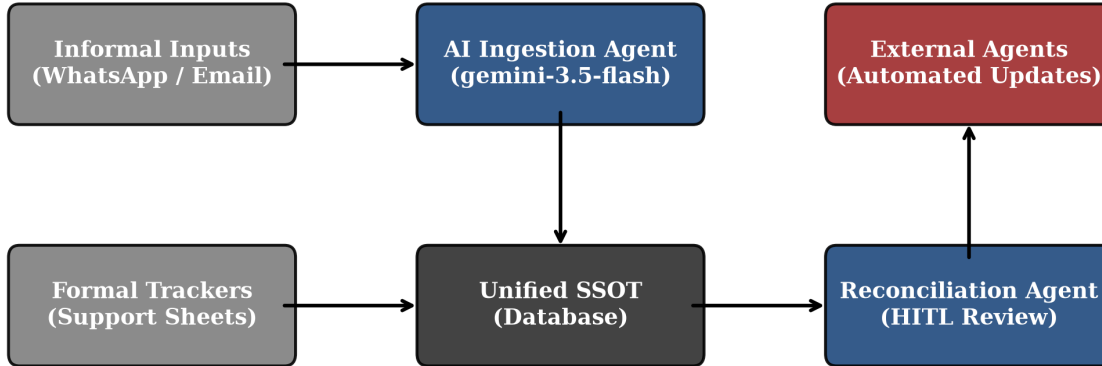


Figure 2: System Architecture Flowchart

exposed to the LLM’s context window.

5. Measuring Success

To ensure the solution is working, we will monitor two primary metrics over the next month:

1. **Escalation Rate:** The percentage of refund requests that result in a formal escalation. **Target: 85%+ drop within 30 days.** Because 149 mismatches + 100 drops + 24 WhatsApp leaks account for virtually all operational friction, successfully deploying this SSOT proactively eliminates the root causes of the vast majority of escalations.
2. **Tracker Discrepancy Count:** The number of tickets marked “Closed” by Support without a matching payout record in Finance. **Target: Zero.**

6. Assumptions Made

As the brief was deliberately incomplete, I proceeded with the following assumptions to build the prototype:

- **Finance is the ultimate authority on payout amounts:** If Support quotes 5000 INR but Finance pays 4500 INR, I assumed the 500 INR deduction is legitimate (cancellation fee) rather than a Finance error, meaning the fix requires better communication to the agent rather than a refund correction.
- **WhatsApp/Email formats are unpredictable:** I assumed agents will not follow a strict template, necessitating an LLM to dynamically extract the `agent_name`, `route`, and intent from informal natural language.